

Daily Review

Booklet p.22-23 — Spectra, and p.15 — $F = ma$

- 1 Name the two types of spectrum.
- 2 How do we know which elements are in a star?
- 3 Which type of spectrum does a filament lamp give?
- 4 A 2.0 kg trolley has an unbalanced force of 6.0 N. Find a .
- 5 What is a line spectrum?

Continuous and line

Match its lines to known ones

Continuous

3.0 m s^{-2}

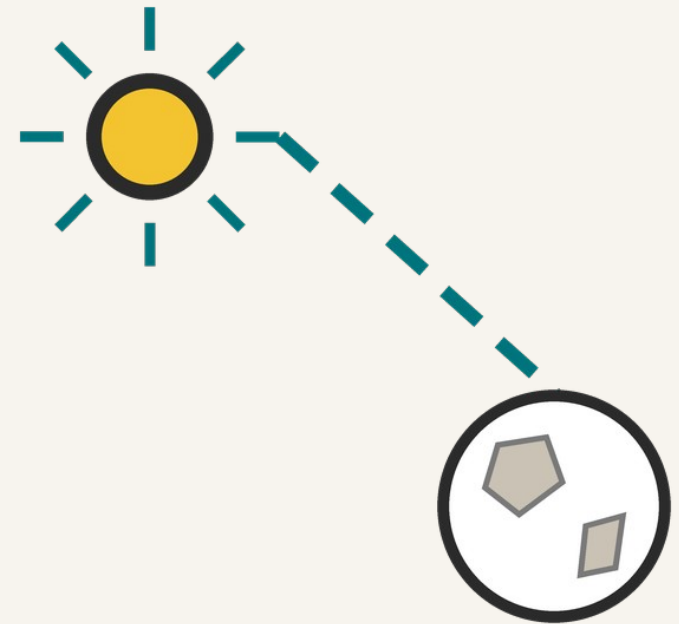
Separate bright lines

SPACE

Light Years

Space is so big that we measure it with light.

National 5 Physics (S3)



THE BIG QUESTION

This is the nearest star to the Sun — and its light still takes 4.2 years to get here. How far is that in metres?



ESA/Hubble & NASA (CC BY 4.0)

PART 1

What a light year is

Why space is measured in light years, and not in metres.

By the end of this part...

- 1 I can correctly use the term light year
- 2 I can explain why distances in space are given in light years

Show me — Describe each word

SHOW ME BOARDS

1 light year

The distance light travels in a year

2 distance

How far apart two things are

3 speed of light

$3.0 \times 10^8 \text{ m s}^{-1}$

4 second

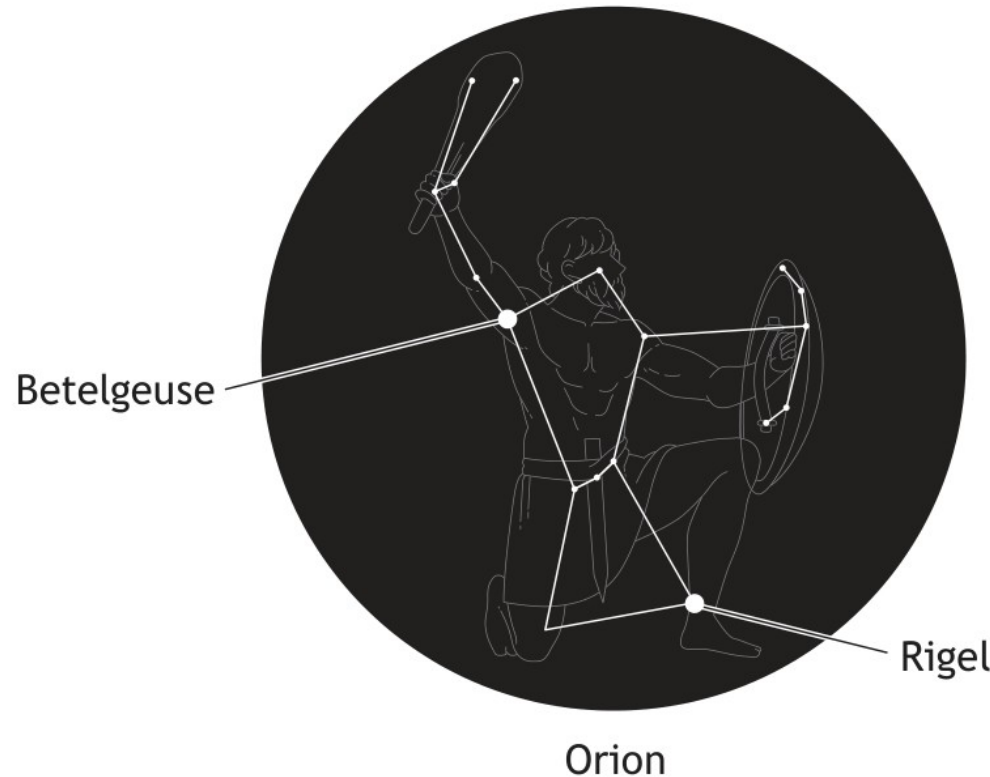
The unit of time (s)

5 scientific notation

Powers of ten, for huge numbers

How far away are they?

Orion

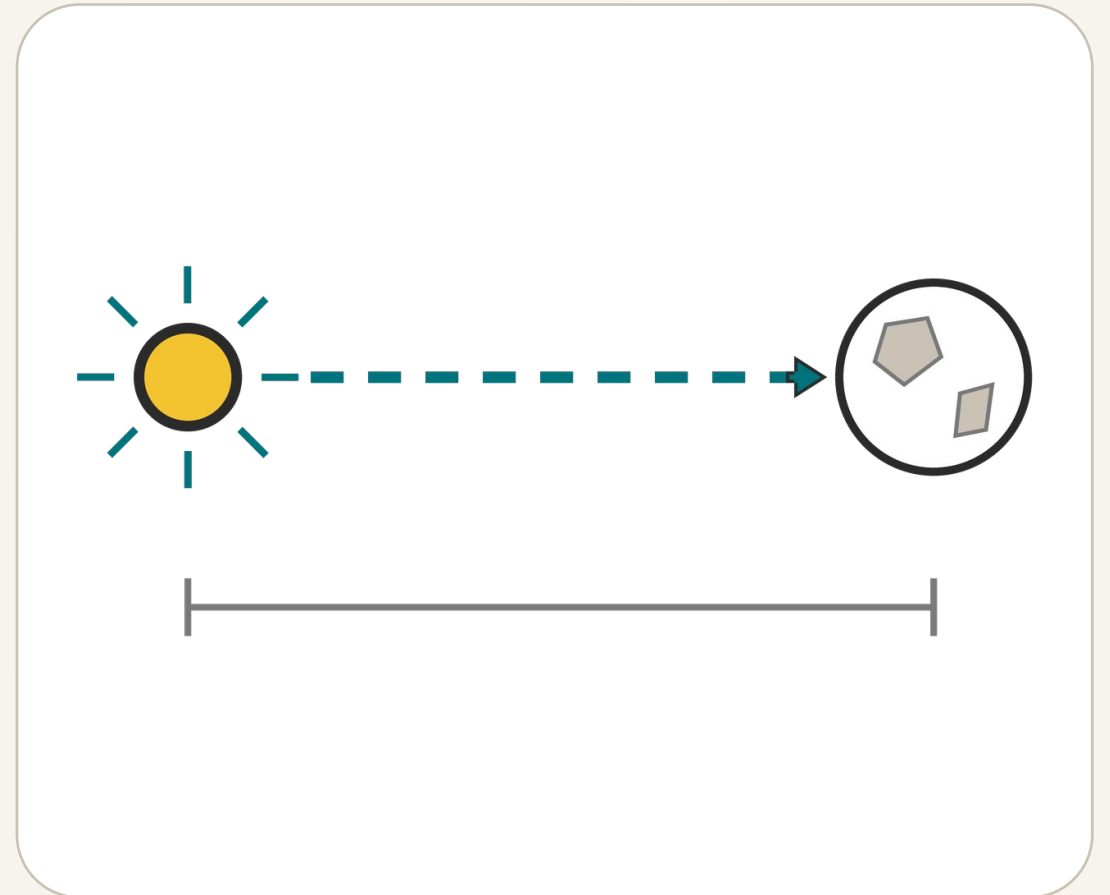


Betelgeuse and Rigel are hundreds of light years away.

What is a light year?

Booklet p.24

- Light travels at $3.0 \times 10^8 \text{ m s}^{-1}$.
- A light year is the distance light travels in one year.
- It is a distance, not a time.



Why not metres?

Booklet p.24

- Distances between stars are enormous.
- In metres the numbers become unmanageable.
- Light years give smaller, more useful numbers.



Task 1 – the light year

Booklet p.24

Complete the definition.

Booklet p.24, Task 1

Task 1 – Answers

ANSWERS

Booklet p.24

- A light year is the distance travelled by light in one year.

One year in seconds

Booklet p.24

- 1 year = 365 days
- 1 day = 24 hours
- 1 hour = 60 minutes, and 1 minute = 60 seconds
- $1 \text{ year} = 365 \times 24 \times 60 \times 60 = 31\,536\,000 \text{ s}$
- $1 \text{ year} = 3.15 \times 10^7 \text{ s}$ — the value on the data sheet

Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 A light year is an amount of time.
- 2 The nearest star is a few million km away.
- 3 Looking at the Sun, we see it as it is now.
- 4 Andromeda looks the way it does today.
- 5 A light year is a distance.

Disagree: it is a distance

Disagree: it is light years away

Disagree: 8 minutes ago

Disagree: as it was long ago

Agree

Daily Review

Booklet p.24 — Part 1

- 1 What is a light year?
- 2 Is a light year a time or a distance?
- 3 How many seconds are there in one year?
- 4 Why not measure in metres?
- 5 State the speed of light.

The distance light travels in a year

A distance

$$3.15 \times 10^7 \text{ s}$$

The numbers are far too big

$$3.0 \times 10^8 \text{ m s}^{-1}$$

PART 2

Working with light years

One relationship, and one conversion, done the same way every time.

By the end of this part...

- 1 I can convert between light years and metres
- 2 I can set the working out the way the exam expects

The relationship

Booklet p.24 – Task 2

$$d = v t$$

d

distance, in
metres (m)

v

speed, in m s^{-1}

t

time, in seconds
(s)

Task 2 – label the equation

Booklet p.24

Label each letter in $d = vt$
with its quantity and its unit.

Booklet p.24, Task 2

Task 2 – Answers

ANSWERS

Booklet p.24

Letter	Quantity	Unit
d	distance	metre (m)
v	speed	m s^{-1}
t	time	second (s)

Worked example 1

Light years to metres

A star is 12 light years from Earth.
Calculate this distance in metres.

$$v = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$t = 12 \text{ years}$$

$$d = ?$$

$$t = 12 \times 3.15 \times 10^7 = 3.78 \times 10^8 \text{ s}$$

$$d = v t$$

$$d = 3.0 \times 10^8 \times 3.78 \times 10^8$$

$$\underline{d = 1.1 \times 10^{17} \text{ m}}$$

Your turn 1 — a star 8.0 light years away

SHOW ME BOARDS

A star is 8.0 light years from Earth.
Calculate this distance in metres.

$$v = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$t = 8.0 \text{ years}$$

$$d = ?$$

$$t = 2.52 \times 10^8 \text{ s}$$

$$d = 7.6 \times 10^{16} \text{ m}$$

$$t = 8.0 \times 3.15 \times 10^7 = \dots\dots \text{ s}$$

$$d = v t$$

$$d = 3.0 \times 10^8 \times \dots\dots$$

$$\underline{d = \dots\dots \text{ m}}$$

Worked example 2 – back to light years

Light from a star travels 2.84×10^{17} m to reach Earth.
How far away is the star, in light years?

$$d = 2.84 \times 10^{17} \text{ m}$$

$$v = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$t = ?$$

$$t = d/v$$

$$t = 2.84 \times 10^{17} / 3.0 \times 10^8$$

$$t = 9.47 \times 10^8 \text{ s}$$

$$t = 9.47 \times 10^8 \div 3.15 \times 10^7 = 30 \text{ years}$$

$$\underline{d = 30 \text{ light years}}$$

Your turn 2 – back to light years

SHOW ME BOARDS

Light travels 5.67×10^{17} m from a star to Earth.
How far away is the star, in light years?

$$d = 5.67 \times 10^{17} \text{ m}$$

$$v = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$t = ?$$

$$t = 1.89 \times 10^9 \text{ s}$$
$$d = 60 \text{ light years}$$

$$t = d/v$$

$$t = \dots\dots \text{ s}$$

$$t = \dots\dots \div 3.15 \times 10^7$$

$$\underline{d = \dots\dots \text{ light years}}$$

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A star is 6.0 light years from Earth.
Its distance from Earth, in metres, is

- A $1.8 \times 10^9 \text{ m}$
- B $1.9 \times 10^8 \text{ m}$
- C $5.7 \times 10^{16} \text{ m}$
- D $9.45 \times 10^{15} \text{ m}$
- E $5.7 \times 10^{15} \text{ m.}$

Answer C

Show me — light years

SHOW ME BOARDS

EXTRA PRACTICE

1 How far does light travel in one year?

$9.45 \times 10^{15} \text{ m}$

2 A star is 10 light years away. Distance in metres?

$9.45 \times 10^{16} \text{ m}$

3 A galaxy is $9.45 \times 10^{20} \text{ m}$ away. How many light years?

100 000

4 Write 3.15×10^7 as an ordinary number.

31 500 000

5 Why use light years instead of metres?

Smaller, easier numbers

Correct your booklet

Booklet p.24, Examples

CROSS OUT AND FIX

Proxima Centauri: “t = 4.2 year” should read “t = 4.2 years”.

Examples – lay them out

Booklet p.24

Two conversions: 1 light year, and
Proxima Centauri at 4.2 light years.
Lay them out exactly as in the tests.

Booklet p.24, Examples

Examples – Answers

ANSWERS

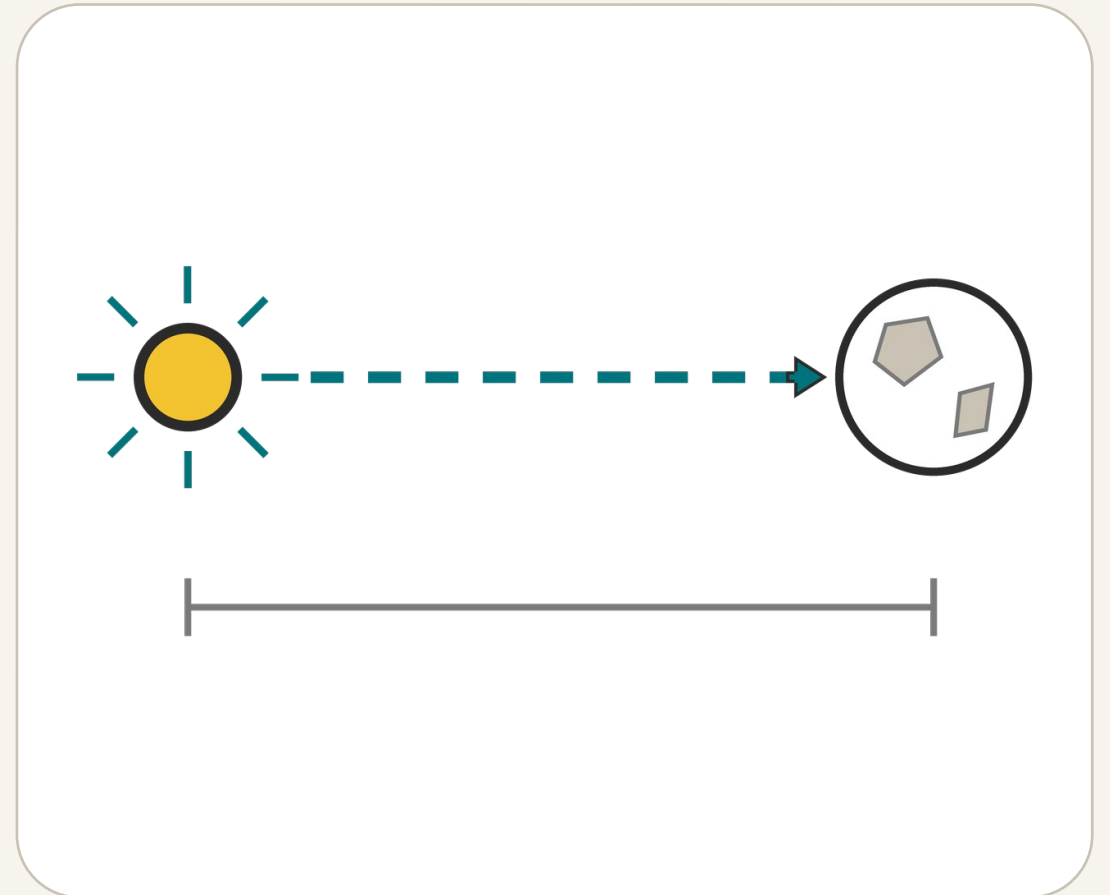
Booklet p.24

	Time, in seconds	Distance, in metres
1 light year	$t = 3.15 \times 10^7 \text{ s}$	$d = 9.45 \times 10^{15} \text{ m}$
4.2 light years	$t = 1.32 \times 10^8 \text{ s}$	$d = 3.97 \times 10^{16} \text{ m}$

Looking back in time

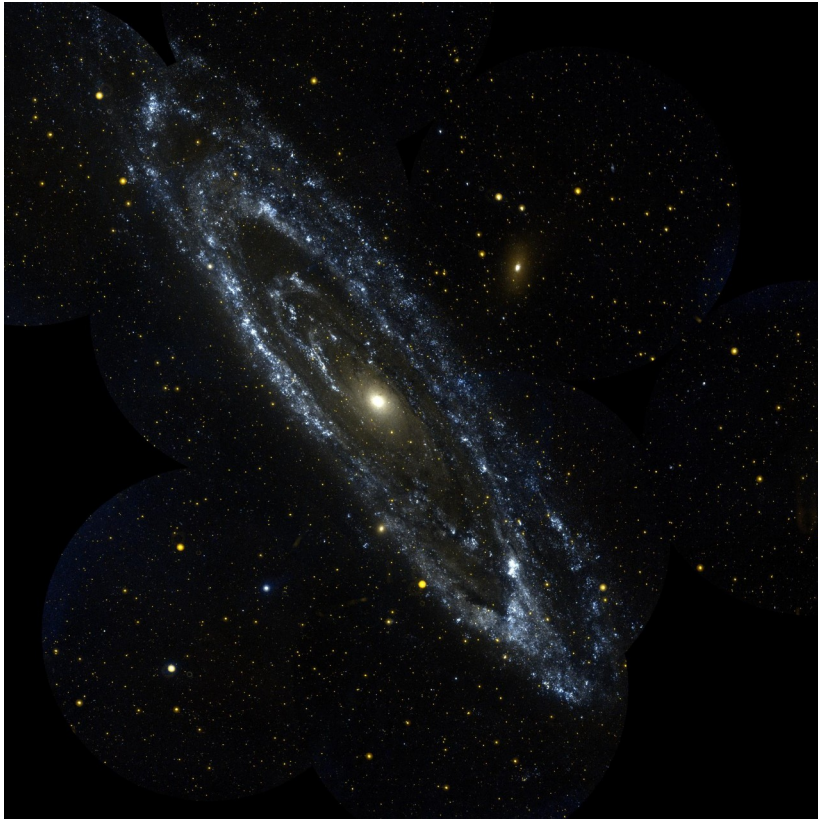
Booklet p.24-25

- Light from a star takes years to reach us.
- We see the star as it was when that light left it.
- A star 100 light years away is seen as it was 100 years ago.



Andromeda, tonight

Booklet p.25, Q3



- Andromeda is 2.5 million light years away.
- Its light set off 2.5 million years ago.
- You are seeing the galaxy as it was then.
- This is an ultraviolet view, from GALEX.

Demo — the corridor Solar System

DEMO

Roll a paper strip along the corridor: 10 m.
Put the Sun at one end and Neptune at the other.
Mark the planets in between, to scale.

On that scale the nearest star is 90 km away.

Extension – BBC Bitesize

Booklet p.25

Revise cosmology on BBC Bitesize.

Booklet p.25, Extension



BBC Bitesize

Checkpoint

- 1 What is a light year?
- 2 A star is 3.0 light years away. Distance in metres?
- 3 Why are distances in space given in light years?

The distance light travels in a year

$$2.8 \times 10^{16} \text{ m}$$

In metres the numbers are huge

Exam practice

Past paper

- National 5 Physics 2024, Section 2, question 6(b)(ii). 3 marks.
- 2023, Section 2, question 5(c). 2 marks. A star's distance in metres.
- Then booklet p.24-25, Task 3.

LOOKING AHEAD

Telescopes

Why we look at the sky in far more than visible light.

Plenary

- 1 The big question: how far is Proxima Centauri, in metres?
- 2 Define a light year.
- 3 State the speed of light in a vacuum.
- 4 How many seconds are in one year?
- 5 A star is 20 light years away. Distance in metres?

$$3.97 \times 10^{16} \text{ m}$$

The distance light travels in a year

$$3.0 \times 10^8 \text{ m s}^{-1}$$

$$3.15 \times 10^7 \text{ s}$$

$$1.89 \times 10^{17} \text{ m}$$

Plenary

EXTRA PRACTICE

- 1 Write the relationship between distance, speed and time.
- 2 Convert 2 years into seconds.
- 3 How far, in metres, is 1 light year?
- 4 A galaxy is 1.89×10^{16} m away. How many light years?
- 5 What do we see, looking at a star 50 light years away?

$$d = vt$$

$$6.3 \times 10^7 \text{ s}$$

$$9.45 \times 10^{15} \text{ m}$$

2

It as it was 50 years ago

Extension

- 1 Sunlight takes about 8 minutes to reach us. Estimate the Sun's distance.
- 2 A star 600 light years away exploded 100 years ago. Can we see it yet?
- 3 How many metres are there in 1000 light years?

Extension

EXTRA PRACTICE

- 1 Moonlight takes about 1.3 s to reach us. How far away is the Moon?
- 2 Why does looking further out in space mean looking further back in time?
- 3 Suggest why a star we can see tonight might no longer exist.

Course specification

N5 Physics course specification v4.0, p.8 – Cosmology

- Use of the term ‘light-year’ and conversion between light-years and metres.